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Work machine with housing and case

Abstract

A work machine that can achieve size reduction and weight reduction of a housing. The work machine includes a housing, a motor fixed to an upper surface of the housing, a battery connected with the motor, and a blade driven by the motor. The motor is covered by a motor case fixed to the upper surface of the housing. The motor case is fixed to the upper surface of the housing in a different part from the motor.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a work machine.

BACKGROUND ART

(2) In related art, a work machine has been known which includes a housing as a working machine main body, a working unit performing predetermined work, a motor as a motive power source of the working unit, and a battery supplying electric power to the motor. Among such work machines, a work machine has been known in which an electric motor and a battery are attached to a housing via a shared attachment member (for example, see Patent Literature 1).

CITATION LIST

Patent Literature

(3) [Patent Literature 1]

(4) Japanese Patent Laid-Open No. 2018-102251

SUMMARY OF INVENTION

Technical Problem

(5) However, in a configuration in related art, because loads of an electric motor and a battery are concentrated in a predetermined attachment part, strength and rigidity of the housing have to be enhanced, and this may lead to a size increase and a weight increase of the housing.

(6) An object of one aspect of the present invention is to provide a work machine that can achieve size reduction and weight reduction of a housing.

Solution to Problem

(7) One aspect of the present invention provides a work machine including: a housing; a motor fixed to an upper surface of the housing; a battery connected with the motor; and a working unit to be driven by the motor, in which the motor is covered by a case fixed to the upper surface of the housing, the battery is fixed to an upper surface of the case covering the motor, and the case is fixed to the upper surface of the housing in a different part from the motor.

(8) In one aspect of the present invention, in the work machine, the case is fixed to a first fixing portion provided to the housing, and the first fixing portion is formed in a wider range than an external shape of the battery.

(9) In one aspect of the present invention, in the work machine, the motor and the battery are arranged side-by-side on the upper surface of the housing, and the first fixing portion is formed in a wider range than a periphery of an external shape of the motor and the external shape of the battery.

(10) In one aspect of the present invention, in the work machine, the housing is provided with a second fixing portion within a range surrounded by the first fixing portion.

(11) In one aspect of the present invention, in the work machine, the case is fastened to the first fixing portion by a fastening member, fitted to the second fixing portion, and thereby fixed to the housing.

(12) In one aspect of the present invention, in the work machine, the case is provided with a cover covering the battery in an openable and closable manner.

(13) In one aspect of the present invention, in the work machine, the battery is provided in front of the motor.

(14) In one aspect of the present invention, in the work machine, the working unit is a blade that mows lawn grass below the housing.

Advantageous Effects of Invention

(15) In one aspect of the present invention, size reduction and weight reduction of a housing can be achieved.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view that illustrates an outline configuration of a work machine according to an embodiment of the present invention.
- (2) FIG. 2 is a perspective view of a housing.
- (3) FIG. 3 is a perspective view of a power unit as seen from a lateral side.
- (4) FIG. 4 is a vertical cross-sectional view in a longitudinal direction, which illustrates an outline configuration of an inside of the power unit.
- (5) FIG. 5 is a perspective view of the power unit as seen from below.
- (6) FIG. 6 is a perspective view that illustrates the power unit with a battery cover detached.
- (7) FIG. 7 is a perspective view of a flat plate portion.
- (8) FIG. 8 is a perspective view that illustrates a motor, an air introduction member, and a control device.
- (9) FIG. 9 is a perspective view of the air introduction member as seen from the rear, which is placed on the motor.

DESCRIPTION OF EMBODIMENT

- (10) An embodiment of the present invention will hereinafter be described with reference to drawings. In this embodiment, a lawn mower will be described as an example of a work machine.
- (11) Note that in the description, as for descriptions of front, rear, left, right, up, and down, the forward travel direction of the work machine is defined as the front of a vehicle unless otherwise mentioned. A reference character FR indicated in each of drawings denotes the front of the vehicle, a reference character UP denotes the upper side of the vehicle, and a reference character LH denotes the left of the vehicle.
- (12) FIG. 1 is a side view that illustrates an outline configuration of a work machine **1** according to the embodiment of the present invention.
- (13) The work machine **1** of this embodiment is a walking lawn mower that moves by being pushed by a worker performing lawn mowing work.
- (14) The work machine **1** includes a housing **10** as a main body. The housing **10** is a member that houses a blade **11** as a working unit of the work machine **1** on an inside, and in the housing **10**, a lower surface facing a lawn ground is open.
- (15) A pair of left and right front wheels **12** are provided in a front portion of the housing **10**, and a pair of left and right rear wheels **13** are provided in a rear portion of the housing **10**. An operation handle **14** extending rear-upward is provided in a rear portion of the housing **10**. An operation unit **15** by which the worker operates rotation, stop, the number of rotations, and so forth of the blade **11** is provided in an upper end portion of the operation handle **14**.
- (16) FIG. 2 is a perspective view of the housing **10**.
- (17) As illustrated in FIG. 2, an insert-through hole **21** for causing an upper surface to communicate with the lower surface is provided at a center of the housing **10**, and plural motor fixing portions **22** formed to protrude upward from the upper surface of the housing **10** are provided around the insert-through hole **21**.
- (18) A first case fixing portion **23** is provided on the upper surface of the housing **10**. The first case fixing portion **23** is formed to protrude upward from the upper surface of the housing **10** to a predetermined height, and the first case fixing portion **23** is formed in an annular shape that surrounds the insert-through hole **21** and each of the motor fixing portions **22** on the upper surface of the housing **10**. That is, a front end side of the first case fixing portion **23** in the front-rear direction of the housing **10** is arranged between the pair of front wheels **12**, and a rear end side is arranged in the rear of each of the motor fixing portions **22**.
- (19) In the first case fixing portion **23**, case fastening portions **24** into which screw members **94** can be threaded are provided at predetermined intervals in the circumferential direction of the first case fixing portion **23**.

(20) On the upper surface of the housing **10**, a pair of second case fixing portions **25** are provided to protrude in a range surrounded by the first case fixing portion **23** and in parts positioned on a front side of the motor fixing portions **22**.

(21) FIG. **3** is a perspective view of a power unit **17**, and FIG. **4** is a vertical cross-sectional view in the longitudinal direction, which illustrates an outline configuration of an inside of the power unit **17**. FIG. **5** is a perspective view of the power unit **17** as seen from below.

(22) The power unit **17** is arranged on the upper surface of the housing **10**. As illustrated in FIG. **3** and FIG. **4**, the power unit **17** is a unit that houses a motor **30** and a battery **40**.

(23) The motor **30** of this embodiment is a drive source for rotating the blade **11**, and the motor **30** is driven by electric power supplied from the battery **40**.

(24) As illustrated in FIG. **5**, the motor **30** is fixed to an upper surface of a motor retainment member **33**. The motor retainment member **33** includes plural retainment holes **34**, each of which passes through from the upper surface to a lower surface. The retainment holes **34** and the respective motor fixing portions **22** are fastened together by fastening members such as bolts in a state where the retainment holes **34** are arranged in the positions corresponding to the respective motor fixing portions **22**, and the motor **30** and the motor retainment member **33** are thereby fixed to the housing **10**.

(25) A motor cover **35** that covers the motor **30** is provided in an outer periphery on the upper surface of the motor retainment member **33**. As illustrated in FIG. **4**, an opening **36** is provided in an upper surface of the motor cover **35**.

(26) Both ends of a driving shaft **37** of the motor **30** respectively protrude from an upper surface and a lower surface of the motor **30**. A lower end side of the driving shaft **37** is inserted through the insert-through hole **21**, and the blade **11** arranged on the lower surface of the housing **10** is attached to a lower end of the driving shaft **37**. The rotation center of the blade **11** is the driving shaft **37**. That is, the driving shaft **37** rotates by driving of the motor **30**, and the blade **11** is thereby rotated and driven.

(27) A cooling fan **38** is provided at an upper end of the driving shaft **37**. The cooling fan **38** is formed into a disk shape and closes the opening **36** of the motor cover **35**.

(28) Note that if the cooling fan **38** demarcates the motor **30** from a space above the motor **30**, the cooling fan **38** may be provided with a predetermined gap from the opening **36**.

(29) Plural vanes **39** are provided on upper and lower surfaces of the cooling fan **38**. The cooling fan **38** rotates in response to rotation and driving of the motor **30**, the vanes **39** provided on a lower surface side of the cooling fan **38** stir air on an inside of the motor cover **35**, causes thermal diffusion, and thereby cools the motor **30**.

(30) The motor **30**, the motor cover **35**, and the cooling fan **38** are covered by a motor case **50**. The motor case **50** of this embodiment is formed of a resin, and the motor case **50** includes an accommodation portion **51** that covers the motor **30** and a flat plate portion **52** on which the battery **40** is arranged.

(31) A lower surface of the accommodation portion **51** is open, and the accommodation portion **51** covers the motor **30** placed on the upper surface of the housing **10**, a control device **80**, and an air introduction member **70**. In a part positioned on a rear side of the accommodation portion **51** in the front-rear direction of the work machine **1**, a discharge port **96** which opens downward is provided.

(32) FIG. **6** is a perspective view that illustrates the power unit **17** with a battery cover **60** detached, and FIG. **7** is a perspective view of the flat plate portion **52**.

(33) As illustrated in FIG. **6** and FIG. **7**, the flat plate portion **52** is arranged on a front side of the accommodation portion **51** in the front-rear direction of the work machine **1**. A partition wall **53** is provided on a front side in a side surface of the accommodation portion **51**, and the flat plate portion **52** is provided at a lower end of the partition wall **53**. The flat plate portion **52** is formed such that an upper surface is inclined downward from the accommodation portion **51** side toward the front side of the work machine **1**.

- (34) Plural ribs **54** extending along the front-rear direction of the work machine **1** are formed to protrude on the upper surface of the flat plate portion **52**. Each of upper sides of the ribs **54** is formed to be generally horizontal. Thus, each of the ribs **54** protrudes higher from the upper surface of the flat plate portion **52** toward the front side of the work machine **1**. A bottom surface of the battery **40** arranged on the flat plate portion **52** is supported by the upper sides of the ribs **54**.
- (35) A terminal block **91** with which terminals of the battery **40** are electrically connected is provided at a center of the lower end of the partition wall **53**, and the battery **40** and the motor **30** are electrically connected together by the terminal block **91**.
- (36) A pair of side walls **55** are provided on the upper surface of the flat plate portion **52**. The pair of side walls **55** respectively cover portions of both side surface of the placed battery **40**, the side surfaces positioned in the width direction of the battery **40**.
- (37) The flat plate portion **52** is provided with plural bottom surface ventilation ports **56A** for causing the upper surface to communicate with a lower surface of the flat plate portion **52**, that is, an outer side to communicate with an inner side of the motor case **50**. Each of the bottom surface ventilation ports **56A** is provided in a part of the flat plate portion **52** which is adjacent to the partition wall **53** side.
- (38) Similarly, each of the side walls **55** is provided with plural side surface ventilation ports **56B** for causing the outer side to communicate with the inner side of the motor case **50**. Each of the side surface ventilation ports **56B** is formed into a slit shape extending in the up-down direction and is continuous with a portion below the flat plate portion **52** via a communication pipe provided on an inside of the side wall **55**.
- (39) Plural groove portions **57** formed by being notched downward are provided in predetermined parts on a front side in a periphery of the flat plate portion **52**. Each of the groove portion **57** is formed to extend from an outer periphery of the flat plate portion **52** toward an inner side of the flat plate portion **52**.
- (40) On a lower surface side of the flat plate portion **52**, a pair of insertion portions **92** in which the second case fixing portions **25** are inserted are provided to protrude downward. The pair of insertion portions **92** are arranged in positions between which the terminal block **91** is interposed along the width direction of the work machine **1**, and each of the pair of insertion portions **92** has a recessed shape recessed downward.
- (41) A case outer peripheral portion **58** extending outward with a predetermined width throughout a whole circumference is provided at a lower end of the motor case **50**.
- (42) The case outer peripheral portion **58** and the first case fixing portion **23** of the housing **10** are formed into generally the same shape in a planar view. A screw hole **59** is provided in each part of the case outer peripheral portion **58** which corresponds to the case fastening portion **24** of the first case fixing portion **23**.
- (43) The case outer peripheral portion **58** of the motor case **50** is placed on an upper end of the first case fixing portion **23**. The screw member **94** is inserted through and fastened to each of the screw holes **59** and each of the case fastening portions **24** in a state where the pair of insertion portions **92** are respectively inserted in the second case fixing portions **25**, and the motor case **50** is thereby fixed to the housing **10**.
- (44) In the motor case **50** fixed in such a manner, as illustrated in FIG. **4**, a space **V** is provided between the flat plate portion **52** and the upper surface of the housing **10**.
- (45) The battery **40** of this embodiment has a rectangular parallelepiped shape and is housed in the flat plate portion **52** such that the longitudinal direction goes along the front-rear direction of the work machine **1**.
- (46) The battery **40** and the flat plate portion **52** are covered with the battery cover **60** formed of a resin.
- (47) In a closed state, the battery cover **60** covers a portion from an upper end of the partition wall **53** to a front end of a peripheral portion of the flat plate portion **52** in the front-rear direction of the

work machine **1** and covers the pair of side walls **55** in the width direction of the work machine **1**. A periphery of the battery cover **60** contacts with the periphery of the flat plate portion **52** and a whole periphery of a front surface of the accommodation portion **51**. A sealing material is provided to the periphery of the battery cover **60** so as to tightly contact with the periphery of the flat plate portion **52** and a periphery of the partition wall **53** when the battery cover **60** is in the closed state. Accordingly, the battery **40** and the flat plate portion **52** are shielded from trash, dust, and so forth on the outside.

(48) The battery cover **60** is attached to the motor case **50** by hinges **93** provided around the upper end of the partition wall **53** on the upper surface of the accommodation portion **51**, and the battery cover **60** is provided to be openable and closable about the hinges **93** as pivots.

(49) The worker exposes the battery **40** by opening the battery cover **60** and replaces the battery **40**. As described above, the battery **40** is partitioned from the motor **30** and the control device **80** by the motor case **50**. Thus, even in a case where the battery cover **60** is opened, the motor **30** and the control device **80** are not exposed to the outside, and entry of trash and dust from the outside is inhibited.

(50) The plural groove portions **57** are provided in predetermined parts in the periphery of the flat plate portion **52**. Thus, an inside of the battery cover **60** may communicate with an outside even if the battery cover **60** is in the closed state.

(51) FIG. **8** is a perspective view that illustrates the motor **30**, the air introduction member **70**, and the control device **80**, and FIG. **9** is a perspective view of the air introduction member **70** as seen from the rear, which is placed on the motor **30**. Note that in FIG. **9**, the control device **80** is omitted.

(52) Next, an inside of the accommodation portion **51** will be described in detail.

(53) As illustrated in FIG. **8** and FIG. **9**, on the inside of the accommodation portion **51**, the air introduction member **70** having a pentagonal frame shape in a top view is placed above the motor **30**. A partition plate **71** is provided in an intermediate area of the air introduction member **70** in the up-down direction.

(54) A communication hole **74** is provided at a center of the partition plate **71**. The communication hole **74** is provided in a part positioned above the vanes **39** on an upper surface side of the cooling fan **38** and causes a space on an upper side to communicate with a space on a lower side of the partition plate **71** provided to the air introduction member **70**. The space on the lower side of the partition plate **71** covers the vanes **39** on the upper surface side of the cooling fan **38**.

(55) On the upper side of the partition plate **71** of the air introduction member **70**, a surface positioned on a front side in the front-rear direction of the work machine **1** is notched with a predetermined width, and the suction port **73** is thereby formed.

(56) On the lower side of the partition plate **71** of the air introduction member **70**, in a surface positioned on a rear side in the front-rear direction of the work machine **1**, an exhaust port **76** opening rearward is provided.

(57) In such a manner, the air introduction member **70** is provided with an air path through which air flows from the suction port **73** to the exhaust port **76** by rotation of the cooling fan **38**. The upper side of the partition plate **71** forms an upper stage air path **72** of the air introduction member **70**, and the lower side forms a lower stage air path **75** of the air introduction member **70**.

(58) The control device **80** is arranged on an upper surface of the partition plate **71** of the air introduction member **70**. The control device **80** is a member that includes an electric circuit which controls electric power supply performed from the battery **40** to the motor **30**. The control device **80** turns on or off the power supply to the motor **30** in accordance with the operation of the operation unit **15** provided to the operation handle **14** and thereby rotates or stops the motor **30**. The control device **80** changes the number of rotations of the motor **30** in accordance with the operation of the operation unit **15** by the worker.

(59) On a lower surface of the control device **80**, plural heat dissipation fins **81** protruding

downward are aligned so as to extend in the front-rear direction of the work machine **1**. The heat dissipation fins **81** are placed on the upper surface of the partition plate **71**, and the control device **80** is thereby housed in the air introduction member **70**. One end portion of each of the heat dissipation fins **81** is exposed from the suction port **73** of the air introduction member **70**.

(60) In such a manner, the partition plate **71** is provided to the air introduction member **70**, the control device **80** and the motor **30** may thereby be arranged at a predetermined interval, and heat transmission from the motor **30** to the control device **80** may be inhibited.

(61) When the cooling fan **38** is rotated and driven, the air taken from the outside of the work machine **1** into the inside of the battery cover **60** is taken into an inside of the motor case **50** through the bottom surface ventilation ports **56A** and the side surface ventilation ports **56B**. The bottom surface ventilation ports **56A** and the side surface ventilation ports **56B** are positioned below the suction port **73** of the air introduction member **70**, and the air taken in moves upward.

(62) The air taken into the inside of the motor case **50** flows into the air introduction member **70**, and the control device **80** is thereby cooled. This air is exhausted from the air introduction member **70** and thereafter released from the motor case **50**.

(63) A bag-shaped mown lawn grass storage body **16** which stores lawn grass mown by the blade **11** is provided in a rear portion of the housing **10**. The lawn grass mown by the blade **11** is stored in the mown lawn grass storage body **16** while passing through a predetermined path in the housing **10**. Note that the mown lawn grass storage body **16** of this embodiment is detachably provided, and the work machine **1** may be used without the mown lawn grass storage body **16** mounted.

(64) Next, an action of this embodiment will be described.

(65) The worker walks while gripping the operation handle **14** and pushing the work machine **1** forward, the blade **11** driven by the motor **30** rotates, and the work machine **1** thereby mows lawn grass on a lawn ground.

(66) In the work machine **1** of this embodiment, the motor **30** and the battery **40** are arranged side-by-side on the upper surface of the housing **10**. The battery **40** is arranged in front of the motor **30** and extends to a position adjacent to a front end of the housing **10**.

(67) Accordingly, the center of gravity of the work machine **1** can be positioned low, and a work load in a case where the worker performs work by using the work machine **1** can be reduced.

(68) Furthermore, in the upper surface of the housing **10**, a front portion of the housing **10** can effectively be used as a placement portion of the battery **40**. Thus, the load applied to the housing **10** can further be dispersed, and requested strength for the housing **10** can be decreased. Further, the weight of the housing **10** can be decreased, weight reduction can be achieved, and the work load in a case where the worker performs work by using the work machine **1** can be reduced.

(69) In addition, because the battery **40** can be placed in the vicinity of the front end of the housing **10** as described above, the battery **40** can have a large capacity.

(70) Mown lawn grass is accommodated in the mown lawn grass storage body **16** as described above. That is, the weight of the mown lawn grass storage body **16** changes in accordance with progress of work. Even in such a case, because the battery **40** is arranged in a front portion of the housing **10** to the extent that the other end portion of the battery **40** in the longitudinal direction is arranged between the pair of front wheels **12**, balance of the work machine **1** is easily maintained. The mown lawn grass storage body **16** can have a large capacity.

(71) As described above, the motor **30** is attached to the housing **10** via the motor retainment member **33**, and the battery **40** is placed on the flat plate portion **52** of the motor case **50** covering the motor **30**. That is, the motor **30** and the battery **40** are arranged in different positions of the upper surface of the housing **10** and are attached to the upper surface of the housing **10** via different members. Accordingly, parts in which the housing **10** retains the battery **40** and the motor **30** can be dispersed. Thus, the requested strength for the housing **10** can be decreased, the weight of the housing **10** can be decreased, and size reduction and weight reduction can be achieved.

(72) The motor case **50** on which the battery **40** is placed is placed on the first case fixing portion

23. This first case fixing portion 23 is formed around the battery 40 and the motor 30 in the upper surface of the housing 10. That is, the first case fixing portion 23 is formed in a wider range than a periphery of an external shape of the battery 40 and an external shape of the motor 30. Accordingly, the load of the battery 40 can be dispersed in the upper surface of the housing 10. Thus, the requested strength for the housing 10 can be decreased, the weight of the housing 10 can be decreased, and size reduction and weight reduction can be achieved.

(73) Further, the motor case 50 is supported by and fixed to the housing 10 by fitting between the pair of insertion portions 92 and the respective second case fixing portions 25. Accordingly, in a generally central portion of the motor case 50, the motor case 50 can be fixed to the housing 10 without using the screw members 94. Furthermore, the motor case 50 is supported by each of the second case fixing portions 25 in addition to the first case fixing portion 23, and the load of the battery 40 can thereby be dispersed in the upper surface of the housing 10.

(74) The above-described embodiment provides the following effects.

(75) The work machine 1 of this embodiment includes the housing 10, the motor 30 fixed to the upper surface of the housing 10, the battery 40 connected with the motor 30, and the working unit driven by the motor 30. The motor 30 is covered by the motor case 50 fixed to the upper surface of the housing 10, and the battery 40 is fixed to an upper surface of the motor case 50 covering the motor 30. Further, the motor case 50 is configured to be fixed to the upper surface of the housing 10 in a different part from the motor 30.

(76) Accordingly, the parts in which the housing 10 retains the battery 40 and the motor 30 can be dispersed. Thus, the requested strength for the housing 10 can be decreased, the weight of the housing 10 can be decreased, and size reduction and weight reduction can be achieved.

(77) Further, in this embodiment, the motor 30 and the battery 40 are arranged side-by-side on the upper surface of the housing 10, and the motor case 50 is fixed to the housing 10 by the first case fixing portion 23 provided to the housing 10. Further, the first case fixing portion 23 is configured to be formed in a wider range than the external shapes of the battery 40 and the motor 30.

Accordingly, the load of the battery 40 can be dispersed in the upper surface of the housing 10.

(78) Further, in this embodiment, a configuration is made in which the housing 10 is provided with the second case fixing portions 25 within the range surrounded by the first case fixing portion 23. Accordingly, in a generally central portion of the motor case 50, the motor case 50 can be fixed to the housing 10 without using the screw members 94.

(79) Further, in this embodiment, the motor case 50 is configured to be fastened to the first case fixing portion 23 by the screw members 94, fitted to the second case fixing portions 25, and thereby fixed to the housing 10. Accordingly, the motor case 50 is supported by each of the second case fixing portions 25 in addition to the first case fixing portion 23, and the load of the battery 40 can thereby be dispersed in the upper surface of the housing 10. Further, in a generally central portion of the motor case 50, the motor case 50 can be fixed to the housing 10 without using the screw members 94.

(80) Further, in this embodiment, a configuration is made in which the motor case 50 is provided with the battery cover 60 covering the battery 40 in an openable and closable manner. Accordingly, the battery 40 and the flat plate portion 52 are shielded from trash, dust, and so forth on the outside. Further, because the battery 40 is partitioned from the motor 30 and the control device 80 by the motor case 50, even in a case where the battery cover 60 is opened, trash and dust on the outside are inhibited from entering the motor 30 and the control device 80.

(81) Further, in this embodiment, a configuration is made in which the battery 40 is provided in front of the motor 30. Accordingly, the center of gravity of the work machine 1 can be positioned low, the load applied to the housing 10 can further be dispersed, and weight reduction of the work machine 1 can be achieved.

(82) The above-described embodiment represents one aspect of the present invention as an example, and any modification and application are possible without departing from the scope of the

gist of the present invention.

(83) For example, in the above-described embodiment, the first case fixing portion **23** is formed around the battery **40** and the motor **30**. However, the first case fixing portion **23** is not limited to this but may be provided in a narrower range than the periphery of the external shape of the battery **40** and the external shape of the motor **30** as long as the first case fixing portion **23** is provided in a wider range than the external shape of the battery **40**.

(84) Further, for example, in the above-described embodiment, the motor **30** and the battery **40** are arranged side-by-side on the upper surface of the housing **10**. However, arrangement is not limited to this, but such as a case where at least one end portion of the battery **40** is arranged above the motor **30**, a portion of the battery **40** may be arranged to overlap with the motor **30** in a top view of the housing **10**.

(85) Further, for example, in the above-described embodiment, the flat plate portion **52** is provided on the front side of the accommodation portion **51** in the front-rear direction of the work machine **1**, and the battery **40** is arranged on the flat plate portion **52**. However, arrangement is not limited to this, but if the housing **10** has a sufficient space, the flat plate portion **52** and the battery **40** may be arranged in the rear of or on a lateral side of the accommodation portion **51**.

(86) Further, for example, the work machine **1** may be a walking autonomous traveling work machine that includes the motor **30** as a motive power source for travel for driving the pair of front wheels **12** or the pair of rear wheels **13**, the walking autonomous traveling work machine with which the worker walks while driving the work machine **1** autonomously traveling by the motor **30**.

(87) Further, for example, the work machine **1** may be an autonomous traveling work machine that mows lawn grass while performing unmanned autonomous travel.

(88) Further, for example, the work machine **1** may include a so-called hybrid system that is equipped with an engine (internal combustion engine), charges the battery **40** by driving the engine, and drives the motor **30** by charged electric power.

Claims

1. A work machine comprising: a housing; a motor fixed to an upper surface of the housing; a battery connected with the motor; and a working unit to be driven by the motor, wherein the motor is covered by a case fixed to the upper surface of the housing, the battery is fixed to an upper surface of the case covering the motor, the case is fixed to the upper surface of the housing in a different part from the motor, the case is fixed to a first fixing portion provided by the housing, the motor and the battery are arranged side-by-side in a predetermined direction on the upper surface of the housing, the first fixing portion is formed in a wider range than a periphery of an external shape of the motor and the external shape of the battery, a case outer peripheral portion extending outward with a predetermined width throughout a whole circumference is provided at a lower end of the case, the case outer peripheral portion and the first fixing portion are formed into a same shape in a planar view, the case outer peripheral portion is placed on an upper end of the first fixing portion, the housing is provided with a second fixing portion within a range surrounded by the first fixing portion, and the case is fastened to the first fixing portion by a fastening member, fitted to the second fixing portion, and thereby fixed to the housing.

2. The work machine according to claim 1, wherein the case is provided with a cover covering the battery in an openable and closable manner.

3. The work machine according to claim 1, wherein the battery is provided in front of the motor.

4. The work machine according to claim 1, wherein the working unit is a blade that mows lawn grass below the housing.

5. The work machine according to claim 2, wherein the battery is provided in front of the motor.
